

Effect of low-temperature rolling on the tensile behavior of commercially pure tungsten

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Received 17 October 2007; accepted 2 January 2008

Abstract

We have evaluated the tensile behavior of commercially pure tungsten (W) as a function of low-temperature rolling. It is observed that rolling below the nominal recrystallization temperature 1523 K (1250 °C) concomitantly enhances the ductility and strength of W. Strain-rate jump tests and stress-relaxation experiments show that low-temperature rolling also renders reduced strain-rate sensitivity and activation volumes associated with the plastic deformation of W. For W samples rolled at 873 K (600 °C), with a total equivalent strain of ~ 2.5 (including a previous rolling strain of 1.7 introduced at 1073 K [800 °C]), the activation volume for plastic deformation is around $10b^3$ (b is the magnitude of the Burgers vector of W). This is in agreement with a double-kink mechanism with the spreading between the kinks controlled by the forest dislocation density. Fractographic observations indicate that the rolled W exhibits a laminar structure. The layer thickness is a function of rolling temperature, rather than a function of rolling strain introduced.

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Keywords: Rolling; Tungsten; Tensile testing; Ductility; Strain-rate sensitivity; Activation volume

1. Introduction

Tungsten (W) is one of the refractory metals in the body-centered cubic (BCC) family; its melting point is one of the highest among metals (3695 K [3422 °C]) [1]. Tungsten is also one of the few metals that have a very high mass density (19.25 gm/cm³) [1]. Such properties allow W to be used for a great number of applications. Because of a combination of high mass density and high strength, a critical, long sought-after application of W and tungsten-based heavy alloys (WHAs) is to replace depleted uranium (DU) kinetic energy (KE) penetrators [2–6]. Tremendous effort has been invested toward this objective in the past few decades with very limited success [7]. The primary challenge may be attributed to the ceramic-like quasi-

brittle behavior of conventional coarse-grained W produced by powder metallurgy (PM). Tungsten, like many other BCC refractory metals, is strongly sensitive to the small-atom interstitial impurities such as O, C, N, etc., which tend to segregate along the grain boundaries (GBs). Their accumulation weakens the GBs, leading to inter-granular failure when the material is under a mechanical load (especially in tension) [1,8–10]. Recent investigations have demonstrated that severe plastic deformation (SPD) at low temperatures can refine the grain size of W [11], and improve its ductility significantly [4–6]. It is envisioned that innovative nanoengineering is opening a new era for W [12].

Despite a few interesting findings, a systematic study of the various aspects of plastic deformation of SPD processed ultra-fine grained (UFG) W is still lacking. In this paper, we take on the effort to investigate the room-temperature tensile behavior of commercially pure W, rolled at progressively lower temperatures (that is, temperatures far below the recrystallization temperature of W). We examine the yield strength, tensile ductility, strain hardening behavior and particularly, the strain-rate

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dependence of yield strength or flow stress through strain-rate jump and stress-relaxation test. The tensile failure mode is evaluated by detailed scanning electron microscopic observations of the fracture surface of the specimens.

2. Experimental procedures

The starting material for cold rolling was a W slab of commercial purity purchased from Alfa-Aesar Company (Ward Hill, MA). The original dimensions of the W piece were 3.7 mm × 6.4 mm × 25 mm. The nominal purity level was 99.9% (metal basis). The polished sample was etched by standard Murakami's solution to reveal the grain boundaries. The grain size of the un-processed W was around 40 μm. Fig. 1 shows an optical micrograph of the grains in the starting W.

Rolling was performed on a laboratory roll miller. Since W starts to oxidize in air from 873 K (600 °C) [1], all of the specimens were sealed in a stainless steel canister to reduce oxidation of the work piece. The following sequence of rolling conditions was applied to the different work pieces.

Initially, the work piece was rolled at 1073 K (800 °C) from the starting height (6.4 mm) to a thickness of ~1.2 mm (the equivalent strain corresponding to this thickness reduction is ~1.7). Then the rolled W was retrieved from the stainless steel canister, and cut into small pieces for further processing. Subsequent rolling of these pieces reduced the thickness to ~100 μm with an additional equivalent strain of ~2.5; Sample A was rolled at 1073 K (800 °C). Sample B was rolled at 873 K (600 °C). Sample C was rolled at 773 K (500 °C). The procedures for Sample D were slightly different; the initially 1.2-mm thick W piece was rolled to an intermediate thickness of ~300 μm at 773 K (500 °C). The additional equivalent strain introduced during this step was ~1.4. Then the rolling temperature was raised back to 1073 K (800 °C) to roll the work piece to its final thickness of ~100 μm with an additional equivalent strain of ~1.1. Sample F was rolled at 673 K (400 °C). Finally, the original unrolled W was designated as Sample E.

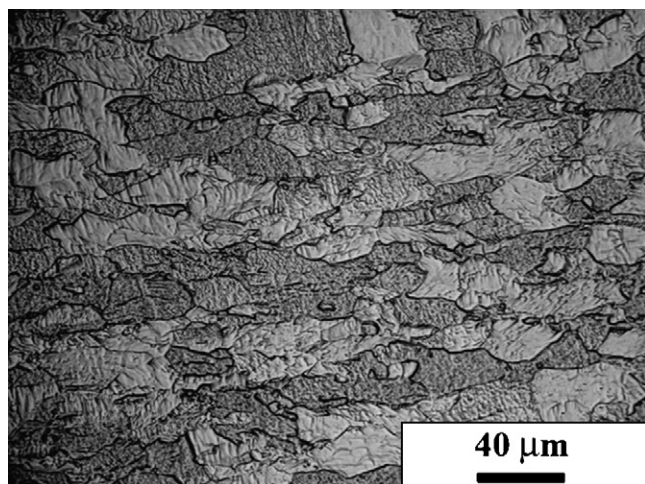


Fig. 1. Optical micrograph of the original coarse-grained W. The average grain size is around 40 μm.

The hardness of the rolled W thin plates was measured using a microhardness tester. Dog-bone type tensile specimens were machined from the thin plates using wire electrical discharge machining (EDM). The gage length of the tensile specimens was 1.5 mm, and the gage width was either 0.75 mm or 1.5 mm. The thickness of the tensile specimens was around 100 μm. Specimens were polished to a mirror finish before tensile testing. Tensile testing was conducted on an Instron 5582 tensile tester with a step-motor screw-driven load system in displacement control mode. The strain rate was set by the speed of the cross-head of the load frame. Stress-relaxation experiments were performed by simply arresting the cross-head and the data acquisition system recorded the load as a function of time. Strain-rate jump tests were performed by a sudden increase of the cross-head speed. For comparison, control samples, machined from the un-processed, coarse-grained W were also tested.

Failed specimens were examined by scanning electron microscopy to identify the failure mechanism of the specimens.

3. Results and discussion

Fig. 2 displays an optical micrograph of Sample A which was rolled to its final thickness at 1073 K (800 °C), throughout. The microstructure has been much refined compared to the original material (see Fig. 1 for comparison). Microhardness measurements of this rolled sample gave a value of ~650 kg/mm². Work by Vashi et al. [13], performed on consolidated W with a range of grain size provides the following Hall–Petch relation in terms of Vickers hardness number,

$$H = H_0 + k_{H-P}d^{-1/2}, \quad (1)$$

where H is the Vickers hardness in kg/mm², H_0 corresponds to the hardness of single crystal W (350 kg/mm²), the Hall–Petch coefficient, k_{H-P} , is 10 kg/mm^{3/2} and the grain size d is in mm. According to Eq. (1), the grain size of the sample displayed in Fig. 2 should be around 1 μm. However, it should also be noted that Eq. (1) is based on a PM processed W. In PM-processed materials, especially W, contamination by interstitials

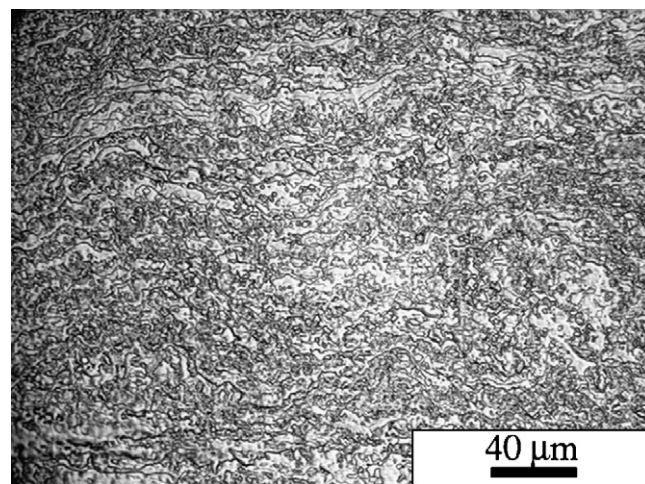


Fig. 2. Optical micrograph of rolled W. The starting rolling temperature is 800 °C and the final rolling temperature is 600 °C, respectively.

is inevitable. As such, they tend to have a higher hardness. Therefore, we should anticipate that Eq. (1) will overestimate the hardness for a given grain size. Conversely, for a given microhardness, due to SPD, the Hall–Petch relation could underestimate the grain size refinement.

The microhardness of Sample B shows a value of $\sim 700 \text{ kg/mm}^2$. This, according to Eq. (1), corresponds to a grain size of about $\sim 800 \text{ nm}$. However, it may be noted that a more accurate determination of the grain size should be obtained via transmission electron microscopy (TEM).

Fig. 3 displays the tensile engineering stress–strain curves of W rolled under the various conditions. The coarse-grained control sample (Sample E) has no ductility at all; it failed at a stress level much smaller than the apparent compressive yield strength ($\sim 1100 \text{ MPa}$ [4,8,14]). In Fig. 3, the failure stress of the coarse-grained control sample is indicated by an X (the letter E represents the barely visible stress–strain curve of the control sample). We point out here that the elastic parts of the curves do not exactly coincide because of certain inaccuracies introduced during the machine compliance correction (in this work no strain gage or extensometer was used to measure the strain; the strain was derived from the displacement of the cross-head).

Fig. 3 shows that low-temperature rolling not only increases the strength of commercially pure W, but also significantly improves the ductility. Early work by Allen et al. and Farrell et al. [10,15] reported the recrystallization temperature of moderately deformed W to be $\sim 1250^\circ\text{C}$. Our highest rolling temperature is

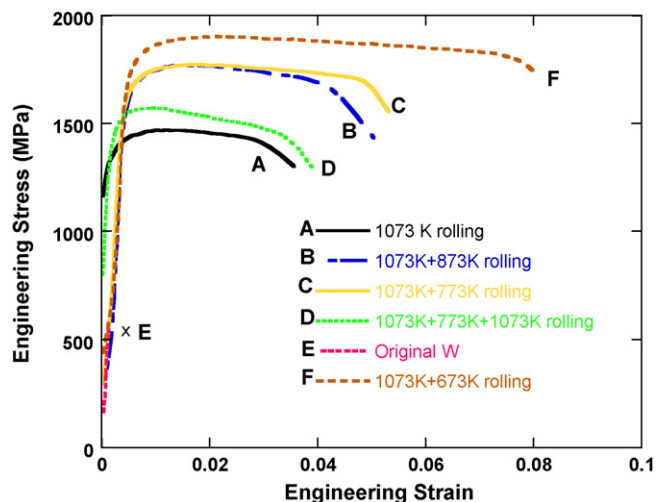


Fig. 3. Engineering stress–strain curves of W rolled at different temperatures. The failure stress of the control coarse-grained sample is indicated by a cross (letter E in the plot). The rolling conditions are also given in the plot.

only 1073 K (800°C), implying that significant dynamic recrystallization might not have occurred during rolling of the W pieces. More careful examination of Fig. 3 suggests that the improvement in ductility increases with decreased rolling temperature. In this study, the 673 K (400°C) rolling to the final thickness gives the highest strength and the best ductility. However, if the final rolling temperature was brought back to the

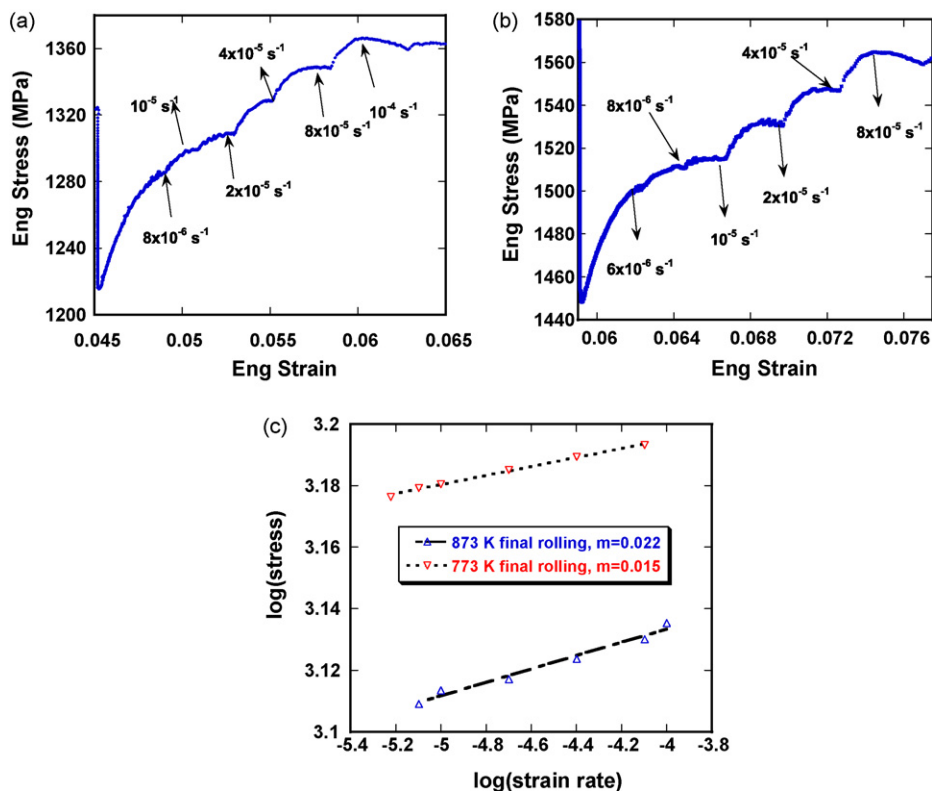


Fig. 4. (a) Engineering stress–strain plot corresponding to a strain-rate jump test of W with final rolling at 873 K (600°C). (b) Engineering stress–strain plot corresponding to a strain-rate jump test of W with final rolling at 773 K (500°C). The strain rate at each jump is indicated in graphs (a) and (b). In (c), the stress versus strain rate is graphed on a double logarithmic scale. The data is from graphs (a) and (b); the strain-rate sensitivities are given in the plot.

original, higher temperature 1073 K (800 °C), such as that shown by Sample D, the benefit gained during the lower temperature rolling is almost diminished (compare Samples A and D and Samples C and D).

Another observation based on the stress–strain curves of Fig. 3 is that beyond the elastic limit, the rolled W exhibits almost no strain hardening, i.e., the engineering stress–strain curves are nearly flat. It may be noted that true strain hardening should be evaluated from the true stress–strain curves. However, for small strains, the true stress and the engineering stress differ from each other only slightly. The lack of strain hardening in the samples, combined with the increasing flow stress and improving ductility with decreasing rolling temperature presents a puzzling phenomenon. It is puzzling because the lack of strain hardening in combination with a high flow stress usually renders an early onset of plastic instability, especially under tension through the necking process [16]. Therefore, the simultaneous increase of strength and ductility in the absence of strain hardening ability needs further explanation.

Fig. 4(a) and (b) displays the engineering stress–strain graphs of two samples, one rolled to the final sample thickness at 873 K (600 °C) and the other at 773 K (500 °C) (Samples B and C, respectively). They correspond to the strain-rate jump experiments. The imposed strain rate corresponding to each jump is

indicated in the plots. The logarithmic flow stress at each strain rate is plotted against the logarithmic strain rate in Fig. 4(c). The strain-rate sensitivity, m , is defined as [17]

$$m = \frac{\partial \ln \sigma}{\partial \ln \dot{\epsilon}}, \quad (2)$$

where $\dot{\epsilon}$ is the imposed strain rate (s^{-1}). Therefore, the strain-rate sensitivity is the slope of the straight line that fits the data points in Fig. 4(c). The strain-rate sensitivity values derived using Eq. (2) are 0.022 and 0.015 for the two rolling conditions, respectively. To confirm the results from the strain-rate jump experiments, we also performed stress-relaxation experiments; Fig. 5(a) shows one example of multiple stress relaxations. In Fig. 5(b) the stress–time curve corresponding to the first relaxation has been re-plotted. The strain-rate sensitivity is calculated using the following equation [18,19]

$$m = \frac{d \ln \sigma}{d \ln \dot{\sigma}} \quad (3)$$

where $\dot{\sigma}$ is the time derivative of the stress (rate of change of stress). Fig. 5(c) displays the logarithmic stress against the logarithmic time derivative of the stress, and the data points corresponding to the early stage of stress relaxation. According to Eq. (3), the slope of the straight line that fits the data is the

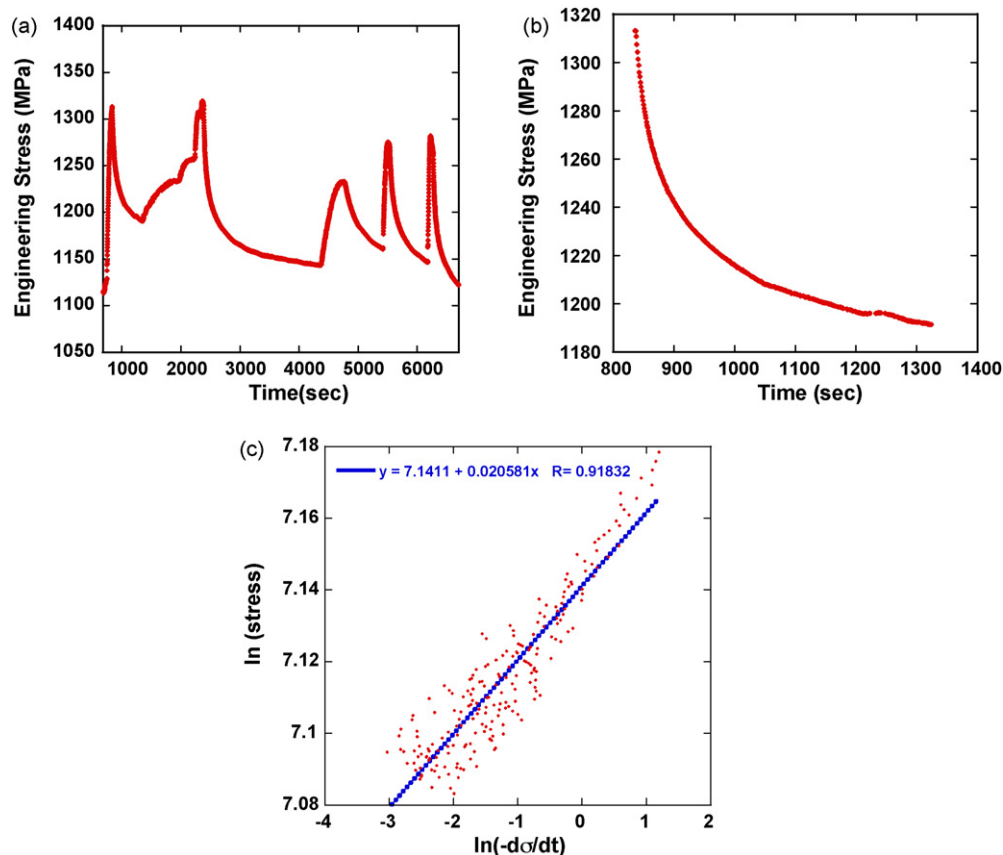


Fig. 5. (a) Engineering stress–time graph showing the multiple stress-relaxation experiments for a W sample rolled under the same condition as that of Sample B, shown in Fig. 4. (b) Engineering stress–time graph of the first stress-relaxation experiment, shown in graph (a). (c) Plot of logarithmic stress versus logarithmic time derivative of stress. The data points correspond to the early stage of relaxation. The strain-rate sensitivity is the slope of the fitted straight line (in [c]). Note, it agrees very well with the strain-rate sensitivity extracted from the strain-rate jump tests (compare results to Fig. 4).

strain-rate sensitivity. For this W sample rolled under the same condition as that shown in Fig. 4(a), i.e., Sample B rolled to final thickness at 873 K (600 °C), we obtained a strain-rate sensitivity of 0.021. This is in close agreement with the result obtained from the strain-rate jump tests. The agreement further verifies that the conclusion that the strain-rate sensitivity values measured by the strain-rate jump and stress-relaxation experiments are equally reliable [20].

Single crystal or coarse-grained W, like other BCC metals with similar microstructures, usually exhibits a strong rate dependence, with strain-rate sensitivities between 0.04 and 0.05 [4,17,21]. Fig. 4(c) shows that the low-temperature rolling has effectively reduced the rate dependence of W. Similar observations in UFG or nanocrystalline BCC metals have been reported for Fe [17,22,23], Ta [24] and V [25] processed by various techniques that include powder consolidation, severe plastic deformation, etc.

In addition to the strain-rate sensitivity, the activation volume for plastic deformation is the second most important parameter that sheds light on the deformation mechanisms of metals [26]. If the plastic deformation is assisted by a thermally activated process, the constitutive response of the metal, in terms of strain rate, can be described by an Arrhenius-type relation given as

$$\dot{\varepsilon} = \dot{\varepsilon}_0 \exp \left(-\frac{\Delta G(\sigma^*)}{kT} \right), \quad (4)$$

where the pre-exponential factor, $\dot{\varepsilon}_0$, includes the number of thermal fluctuations, the average strain per successful thermal fluctuation, and the frequency of the vibration of the elemental unit involved in the thermal activation; k is the Boltzmann constant, T is the absolute temperature and ΔG is the activation energy and is usually a strong function of the stress component σ^* , i.e., the resolved net shear stress to drive the deformation mechanism. This shear stress component is defined as $\sigma^* = \sigma_a - \sigma_\mu$, where σ_a is the applied shear stress and σ_μ is the athermal internal stress stemming from long-range obstacles such as grain boundaries, secondary phases, etc. The activation volume of the thermally assisted plastic deformation is the stress

derivative of the activation energy, and can be written in terms of the strain-rate sensitivity as [17,26]

$$v^* = \frac{kT}{\sigma m}. \quad (5)$$

Using Eq. (5) and the strain-rate sensitivity for the rolled Sample A shown in Fig. 3, for which the final thickness was reached by rolling at 873 K (600 °C), the activation volume for plastic deformation is estimated to be around $10b^3$, b being the magnitude of the Burgers vector of W.

It is known that the rate controlling mechanism at low temperatures (usually for $T/T_m < 0.25$, where T is the temperature of interest and T_m is the melting point of the metal) for plastic deformation in single crystal or coarse-grained BCC metals is the nucleation of double kinks along the screw dislocations. The motion of edge dislocations in BCC metals is athermal, and is at least one order of magnitude faster than that of screw dislocations [27]. Thus the activation volume associated with such processes may be written as

$$v^* = bh\lambda^*, \quad (6)$$

where b is the magnitude of the Burgers vector (2.7394 Å for W), h is the height of the kink which is the same as the Burgers vector [28] and λ^* is the critical distance between the two kinks. The critical distance is a strong function of stress. According to Seeger and Schiller (note, the original publication is in German, and cited in detail in Reference [29]), λ^* can be evaluated as

$$\lambda^* = \left(\frac{\mu bh}{8\pi\sigma} \right)^{1/2}. \quad (7)$$

In Eq. (7), b and h have the same physical meaning as in Eq. (6), μ is the shear modulus (160 GPa for W) and σ is the applied stress. Using the experimental results obtained in this work, the critical distance between two kinks in the rolled W for Sample A was around $8b$. Therefore, the physical activation volume obtained using Eqs. (6) and (7) was very close to that obtained from the strain-rate jump and stress-relaxation tests.

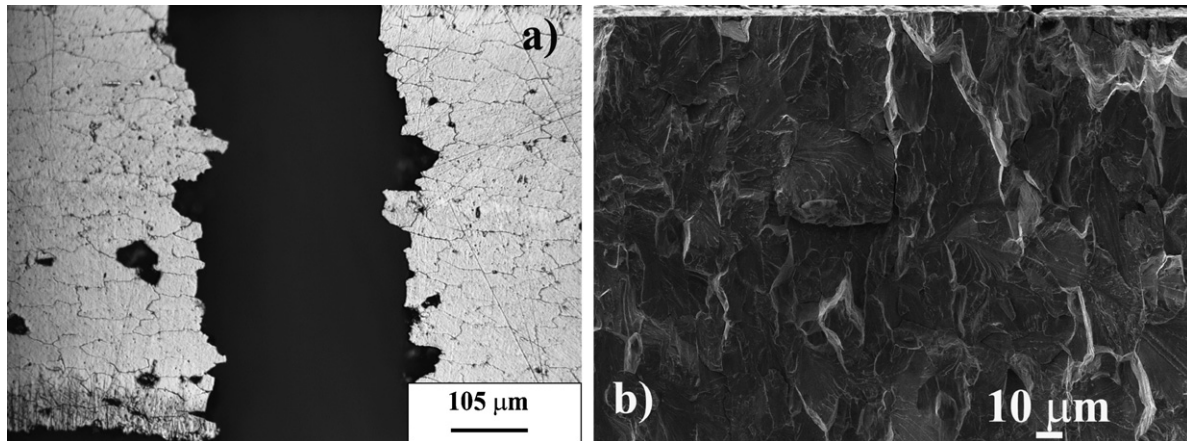


Fig. 6. (a) Optical micrograph of a failed tensile specimen of the original, un-processed W (the two mating parts of the failed specimen are shown in the image). The specimen was polished, but not etched prior to tensile test. However, the grain boundaries are visible after tension, implying very weak grain boundaries that break apart under stress. (b) SEM micrograph showing inter-granular fracture of the original, unrolled W.

According to the early analyses by Dorn and Rajnak [30], the critical spreading between two kinks is around $1\text{ }\mu\text{m}$ for single crystal BCC metals. They also provided evidence that the activation volumes of BCC metals decrease sharply as the effective stress increases and almost reaches a plateau at large stresses, in agreement with the results of many other investigations [31]. Therefore, we believe that Eq. (7) is more suitable for large stress scenarios, such as heavily dislocated metals and UFG (grain size $< 1000\text{ nm}$) and nanocrystalline metals (grain size $< 100\text{ nm}$). In such microstructures, the double-kink spreading is controlled by the distance between the pinning centers such

as forest dislocations, grain boundaries, etc. The very small activation volumes in the rolled W may need further examination. Since the estimated nominal grain size of the rolled W is around $1\text{ }\mu\text{m}$, which is much greater than the experimentally determined

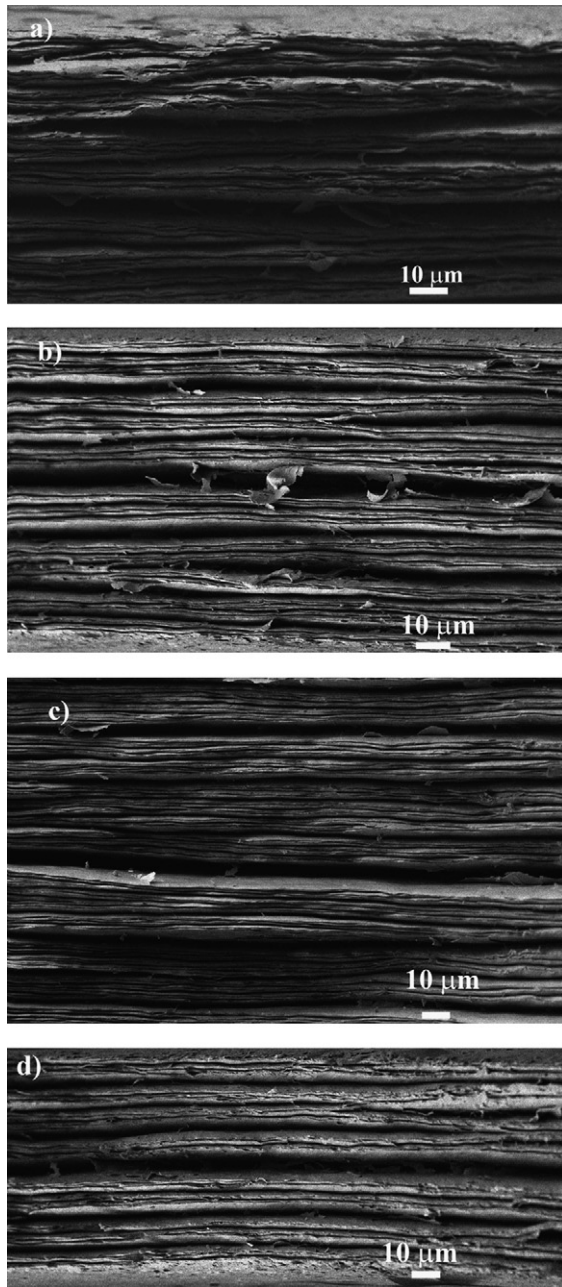


Fig. 7. SEM micrographs of fracture surface of the W samples: Sample A; after rolling at 1073 K (800 °C) in (a), Sample B; 1073 + 873 K (800 + 600 °C) in (b), Sample F; 1073 + 673 K (800 + 400 °C) in (c) and Sample D; 1073 + 773 + 1073 K (800 + 500 + 800 °C) in (d).

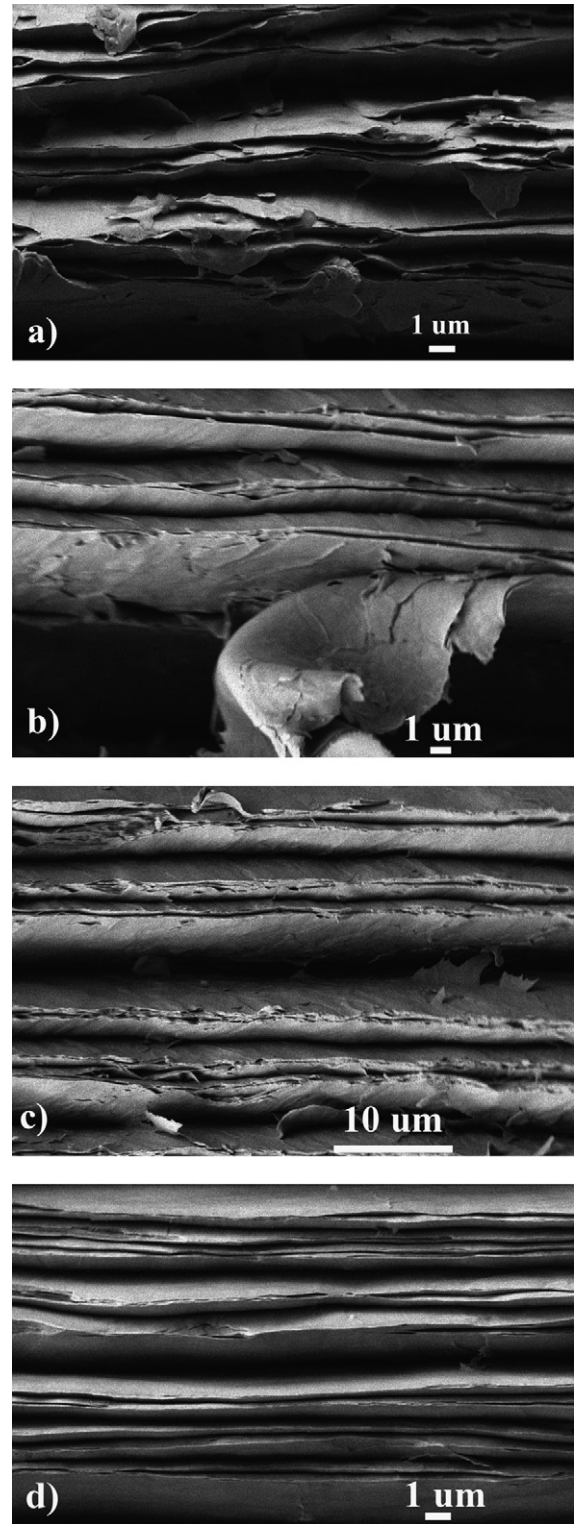


Fig. 8. High-magnification SEM micrographs of the W sample fracture surfaces: Sample A; after rolling at 1073 K (800 °C) in (a), Sample B; 1073 + 873 K (800 + 600 °C) in (b), Sample F; 1073 + 673 K (800 + 400 °C) in (c) and Sample D; 1073 + 773 + 1073 K (800 + 500 + 800 °C) in (d).

double-kink spreading, we presume that it was the dislocations trapped in the grains during severe rolling that gave rise to the small activation volume. To verify this, we took the dislocation density of a heavily deformed crystal to be 10^{17} m^{-2} , the average distance between the dislocations to be about $10b$, which was very close to the double-kink spreading derived from the experiments and Eq. (7). In fact, residual edge dislocations have indeed been observed via high resolution TEM in W severely deformed at low temperature [6].

Fig. 6(a) displays the optical micrograph of the two mating parts of a failed tensile specimen of the original, un-processed W. No sign of plastic deformation was observed. The specimen was polished to mirror finish, but not etched prior to the tensile test. However, grain boundaries had become visible after loading, suggesting that the grain boundaries were so weak that they broke apart under stress (the failure stress is only around 500 MPa). Fig. 6(b) shows an SEM micrograph of the fracture surface of the unrolled W, manifesting typical inter-granular failure mode.

Fig. 7 presents low magnification SEM micrographs of the fracture surfaces of W rolled samples under various conditions. They show that in general the rolled W has a laminated microstructure. The thickness of the laminates, however, is a function of rolling temperature, rather than a function of rolling strain (keep in mind that all the specimens experienced nearly the same total rolling strain). Lower-temperature rolling results in thinner laminates. Higher magnification SEM micrographs in Fig. 8 indicate large plastic deformation during the tensile test. Tearing of the laminates occurs in the samples rolled at higher temperatures (e.g., Figs. 8(a–c)), resulting in the rough appearance of the fracture surfaces under SEM. Samples rolled at a lower temperature show cleaner fracture surfaces. More careful examination of the fracture surface indicates that the laminate fronts have shrunk to a sharp edge. It suggests more uniform plastic deformation of the laminates occurred before failure. This observation may be used to explain the pronounced ductility in the low-temperature rolled W specimens, as was shown in the engineering stress–strain curves in Fig. 3.

4. Summary and concluding remarks

The tensile behavior of low-temperature rolled commercially pure W has been investigated in terms of strength, ductility, failure, strain-rate sensitivity and activation volume associated with plastic deformation. The findings are summarized as follows.

- Low-temperature rolling concomitantly increases the strength and ductility of W. For the same total strain, the ductility of W increases with reduced rolling temperature.
- Both strain-rate jump tests and stress-relaxation experiments consistently indicate a much reduced strain-rate sensitivity of the low-temperature rolled W.
- The activation volume of plastic deformation of the W samples rolled at 873 K (600 °C) was only $10b^3$. This suggests that the double-kink mechanism (with a spreading between the

kinks controlled by the forest dislocation density) was still the controlling mechanism for plastic flow in the low-temperature rolled W.

- Fractographic observations indicate that the rolled W exhibits a laminated structure. The separation distance between laminas was a function of rolling temperature, rather than a function of the rolling strain introduced. Lowering the rolling temperature resulted in further smoothing of the individual W filaments on the fracture surface. Individual grains along the fracture surface also exhibited local shrinkage or necking, suggesting large plastic strain before final failure.

Acknowledgements

The authors would like to thank Profs. K.T. Ramesh and E. Ma (The Johns Hopkins University), Mr. R.J. Dowding, and Drs. L.S. Magness and J.W. McCauley (U.S. Army Research Laboratory) for very helpful discussions. This work was sponsored by the U.S. ARL under the ARMAC-RTP Cooperative Agreement #DAAD19-01-2-0003 through the JHU-CAMCS.

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